

① Find the values of unknown variables in the given parametric equation of curve.

$$x = (t \cos \theta - e^{Mt} \sin(0.3t) \sin(\theta) + x)$$

$$y = (42 + t \sin \theta + e^{Mt} \sin(0.3t) \cos(\theta))$$

unknown are

$$\theta, M, x$$

Range

$$0^\circ < \theta < 50^\circ$$

$$-0.05 < M < 0.05$$

$$0 < x < 100$$

$$t \quad 5 < t < 60$$

Solⁿ: We cannot directly solve this like a normal algebra equation.

Due to

- non linear exponential term
- trigonometric terms $\sin(\theta), \cos(\theta)$.

→ Firstly going to solve x .

From the dataset provided we take the values

$$x_{\min} = 59.65 \quad x_{\max} = 109.23$$

$$y_{\min} = 46.03 \quad y_{\max} = 69.68$$

$$\bar{x} = 83.71$$

$$\bar{y} = 58.26$$

From the eqn we rewrite that

$$x - x = t \cos \theta - e^{Mt} \sin(0.3t) \sin(\theta)$$

$$y - 42 = t \sin \theta + e^{Mt} \sin(0.3t) \cos(\theta)$$

Then

$$x - x = t \cos \theta - A \sin \theta \rightarrow ①$$

$$y - 42 = t \sin \theta + A \cos \theta \rightarrow ②$$

We know

$$6 < t < 60$$

From eq ①

$$\text{mean}(x) = \text{mean}(t \cos \theta - A \sin \theta + X)$$

$$\text{mean}(x) = \text{mean}(t \cos \theta) - \text{mean}(A \sin \theta) + \text{mean}(X)$$

$\theta, X \Rightarrow \text{constant}$

$$\text{mean}(x) = \cos \theta \cdot \text{mean}(t) - \sin \theta (\text{mean}(A)) + X$$

$$\therefore \text{mean}(\sin(x)) \approx 0$$

$$\text{so } \boxed{\text{mean}(A) \approx 0}$$

$$\rightarrow \text{mean}(x) \approx \cos \theta \cdot \text{mean}(t) + X$$

t ranges from 6 $\rightarrow$ 60

$$= \frac{60+6}{2} = 33$$

$$\boxed{\text{mean}(t) = 33}$$

$\rightarrow$ From provided points

$$\text{mean}(x) = 83.71$$

$$83.71 \approx 33 \cos \theta + X$$

$$X \approx 83.71 - 33 \cos \theta$$

θ range $0^\circ < \theta < 50^\circ$

$$\cos(0) = 1$$

$$\cos(50) = 0.64$$

$$0.64 < \cos \theta < 1$$

$$33(0.64) < 33 \cos \theta < 33 \rightarrow \text{multiply } 33$$

$$21.1 < 33 \cos \theta < 33$$

$$X = 83.71 - 33 \cos \theta$$

$$\text{Case 1: } X = 83.71 - 33$$

$$X = 50.71$$

$$\text{Case 2: } X = 83.71 - 21.1$$

$$= 62.61$$

So the x range From

$$50.7 < x < 62.6 \quad | \quad x \approx 57$$

→ For Finding The θ, M, x

Method: IGPR (Inverse Geometric parameter Recovery)

The given nonlinear parametric curve is not treated as an optimization

$$x = t \cos \theta - A \sin \theta + x$$

$$y = t \sin \theta + A \cos \theta + 42$$

Matrix structure discovery

$$\begin{bmatrix} x-x \\ y-42 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} t \\ A \end{bmatrix}$$

From ① & ② eqn multiply $\cos \theta \otimes ①$ $\sin \theta \otimes ②$

$$(x-x) \cos \theta = t \cos^2 \theta - A \sin \theta \cos \theta \rightarrow ③$$

$$(y-42) \sin \theta = t \sin^2 \theta + A \sin \theta \cos \theta \rightarrow ④$$

Add ③ + ④

$$(x-x) \cos \theta + (y-42) \sin \theta = t(\sin^2 \theta + \cos^2 \theta)$$

$$(x-x) \cos \theta + (y-42) \sin \theta = t$$

→ $A = -(x-x) \sin \theta + (y-42) \cos \theta$ ∴ inverse rotation from Matrix

$$A = e^{M|t|} \sin(0.3t)$$

$$\frac{A}{\sin(0.3t)} = e^{M|t|}$$

$$\ln \left(\frac{A}{\sin(0.3t)} \right) = M|t|$$

$$M = \frac{1}{|t|} \ln \left(\frac{A}{\sin(0.3t)} \right)$$

Curve Representation

$$x_{pred} = t \cos \theta - e^{M t} \sin(0.3 t) \sin(\theta) + x$$

$$y_{pred} = 42 + t \sin \theta + e^{M t} \sin(0.3 t) \cos \theta$$

L_1 distance

$$L = \sum |x_{actual} - x_{pred}| + \sum |y_{actual} - y_{pred}|$$

→ This the framework that we need to perform 1500 pairs of values. so with these intelligence we will get approx values.